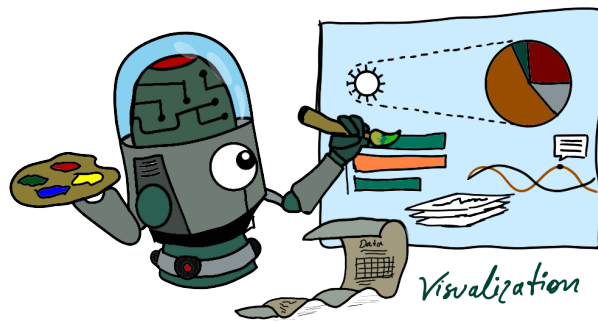


DS5343 - Visualización de Datos



University of Engineering and Technology



Germain Garcia-Zanabria

17 de junio de 2026

Assignment: Visual Learning Models

Goal:

Design visualization that uncovers how to learn a neural network model. You should focus on a clear representation of a neural network model's epochs and layers evolution through the training process. Accompany your visualization with animation resource using the visualization mantra.

For that, you could use the relationships between alternative, high-dimensional representations of observations learned by ANNs and the relationships between artificial neurons within a hidden layer.

In summary, you have to build two visualizations to accomplish the next tasks.

- T1: Exploring the relationships between multidimensional projection of observations learned by ANNs - Considering different EPOCHS.
- T2: Exploring the relationships between multidimensional projection of observations learned by ANNs - Considering different LAYERS.

Use the Figure 1 as inspiration. These visualizations were presented as part of a [PAPER](#)

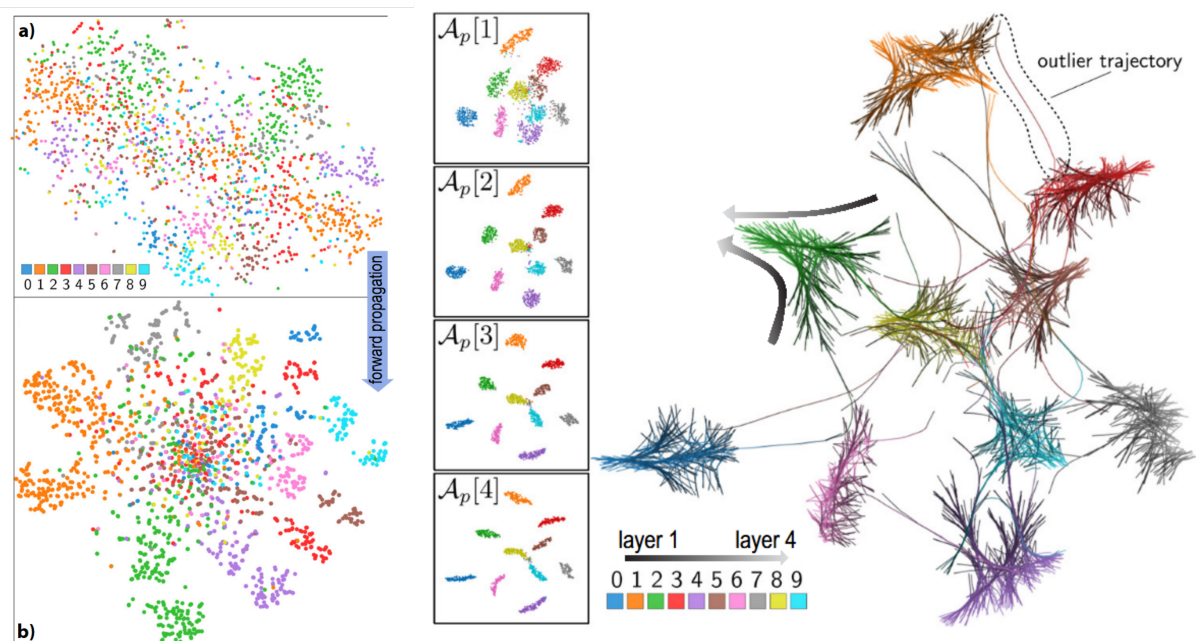


Figure 1: Training Evolution, inter-layer and inter-epochs

Data:

In this activity, you will use a dataset suitable for pattern classification, you could use the same dataset used in the example paper:

- MNIST (handwritten digits)

- SVHN (house number digits)
- CIFAR-10 (object photographs)

These datasets consist of observations (images as high-dimensional vectors) paired with target class assignments. You will need to train an ANN model on this dataset and extract the high-dimensional activations from its hidden layers for a subset of observations (e.g., 1000 test observations). These activations serve as the 'learned representations' of the input data at a specific layer.

Assignment

Your assignment is to implement and apply the visualization techniques, focusing on Tasks T1 and T2 of

- Choose one of the mentioned datasets or a similar image classification dataset.
- Train an ANN model on your chosen dataset.
- Extract the hidden layer activations for a subset of observations (e.g., from the test set) before, during and after training. Consider extracting activations from different layers to explore inter-layer differences.
- Apply a dimensionality reduction technique, such as t-SNE, to the extracted hidden layer activations to project them into a 2D space. The paper used a variation of t-SNE.
- Create scatterplots of the 2D projections.
- Color the points in the scatterplot according to their true class assignments.
- Explore projections from different layers or training stages (e.g., before and after training).
- Implement a method to visualize the evolution of learned representations across layers (inter-layer) or across training epochs (inter-epoch). The paper uses trail bundling for inter-layer evolution visualization

Evaluation Criteria

This assignment is worth 20 points. I will evaluate both the strength of your design and the quality of your computing process. I will also consider factors such as the target audience, the clarity of your message, and how well the intended task is addressed. Your work will be grade based on:

- **Clarity:** Is the visualization easy to interpret?
- **Analytical Insight:** Does it reveal top collaborators and bridges?
- **Design Rationale:** Are your choices well-motivated?
- **Technical Accuracy:** Are network measures calculated correctly?

- **Tasks:** The visualization help to answer 4 tasks effectively.
- **Interaction:** It is mandatory to use interaction and animation resources such as: mouseenter, mouseover, mouseclick, and so on.
- **Number of datasets:** Using more than one dataset will positively impact your final grade

The effectiveness of the projections in representing the high-dimensional data structure. This can be assessed qualitatively by visual inspection (e.g., clarity of clusters, separation)

Deliverables instructions

- Your coded solutions must use D3. Other visualization libraries are not allowed.
- Submit to canvas a single ZIP containing a flask application and data files (nodes and edges).
- You are required to present your work in class. Failure to present will result in no grade for this assignment.

AI support

In this assignment, AI support is allowed.